AIML4002 - INTRODUCTION TO LLM

PRACTICAL 1 - 23DIT024

AIM - Implement an introductory, step-by-step NLP text processing pipeline featuring structural sentence chunking, POS tagging, and entity extraction using spaCy.

```
!pip install -q spacy
!python -m spacy download en_core_web_sm
```

```
Collecting en-core-web-sm==3.8.0
  Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0-py3-none
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 12.8/12.8 MB 147.0 MB/s eta 0:00:00
✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in
order to load all the package's dependencies. You can do this by selecting the
'Restart kernel' or 'Restart runtime' option.
```

Import Required Libraries

We import spaCy and load the English language model.

```
import spacy

# Load English NLP model
nlp = spacy.load("en_core_web_sm")
```

Input Text

We define a paragraph that contains multiple sentences. This text will be processed through the NLP pipeline.

```
text = """
Elon Musk founded SpaceX in 2002.
He is also the CEO of Tesla.
The company headquarters is located in Texas.
On July 15, 2025, Tesla announced a new AI model.
"""
doc = nlp(text)
```

Step 1: Structural Sentence Chunking

Sentence chunking (Sentence Segmentation) divides a paragraph into individual sentences. spaCy automatically detects sentence boundaries.

```
print("Sentence Segmentation:\n")

for i, sent in enumerate(doc.sents, start=1):
    print(f"Sentence {i}: {sent}")
```

```
Sentence Segmentation:

Sentence 1:
Elon Musk founded SpaceX in 2002.

Sentence 2: He is also the CEO of Tesla.

Sentence 3: The company headquarters is located in Texas.

Sentence 4: On July 15, 2025, Tesla announced a new AI model.
```

Step 2: Tokenization

Tokenization is the process of breaking a sentence into individual words, numbers, punctuation marks, and symbols called Tokens.

```
print("Tokens:\n")
for token in doc:
    print(token.text)
```

```
Tokens:



Elon
Musk
founded
SpaceX
in
2002
.


He
is
also
the
CEO
of
Tesla
.


The
company
headquarters
is
located
in
Texas
.


On
July
15
,
2025
,
Tesla
announced
a
new
AI
model
.
```

Step 3: Basic Token Information

spaCy stores useful information about every token. Attributes used:

- text
- lemma
- POS
- shape
- is_alpha
- is_stop

```
print(f"{'Token':15} {'Lemma':15} {'POS':10} {'Shape':10} {'Stop Word'}")
print("-"*65)
for token in doc:
    print(f"{token.text:15} {token.lemma_:15} {token.pos_:10} {token.shape_:10} {token.is_stop}")
```

```
Token           Lemma           POS        Shape      Stop Word
-----------------------------------------------------------------


              SPACE
         False
Elon            Elon            PROPN      Xxxx       False
```

```
Musk            Musk            PROPN       Xxxx        False
founded         found           VERB        xxxx        False
SpaceX          SpaceX          PROPN       XxxxxX      False
in              in              ADP         xx          True
2002            2002            NUM         dddd        False
.               .               PUNCT       .           False


              SPACE
          False
He              he              PRON        Xx          True
is              be              AUX         xx          True
also            also            ADV         xxxx        True
the             the             DET         xxx         True
CEO             ceo             NOUN        XXX         False
of              of              ADP         xx          True
Tesla           Tesla           PROPN       Xxxxx       False
.               .               PUNCT       .           False


              SPACE
          False
The             the             DET         Xxx         True
company         company         NOUN        xxxx        False
headquarters    headquarters    NOUN        xxxx        False
is              be              AUX         xx          True
located         locate          VERB        xxxx        False
in              in              ADP         xx          True
Texas           Texas           PROPN       Xxxxx       False
.               .               PUNCT       .           False


              SPACE
          False
On              on              ADP         Xx          True
July            July            PROPN       Xxxx        False
15              15              NUM         dd          False
,               ,               PUNCT       ,           False
2025            2025            NUM         dddd        False
,               ,               PUNCT       ,           False
Tesla           Tesla           PROPN       Xxxxx       False
announced       announce        VERB        xxxx        False
a               a               DET         x           True
new             new             ADJ         xxx         False
AI              AI              PROPN       XX          False
model           model           NOUN        xxxx        False
.               .               PUNCT       .           False


              SPACE
          False
```

Step 4: Part-of-Speech (POS) Tagging

POS tagging assigns grammatical labels to each word. Common POS Tags:

- NOUN
- VERB
- ADJ
- ADV
- PRON
- PROPN
- DET
- ADP
- AUX

```
print(f"{'Word':15} {'POS':15} {'Explanation'}")

print("-"*55)

for token in doc:
    print(f"{token.text:15} {token.pos_:15} {spacy.explain(token.pos_)}")
```

```
Word            POS             Explanation
-------------------------------------------------------

              SPACE           space
Elon            PROPN           proper noun
Musk            PROPN           proper noun
founded         VERB            verb
SpaceX          PROPN           proper noun
in              ADP             adposition
2002            NUM             numeral
```

```
.               PUNCT          punctuation

                SPACE          space
He              PRON           pronoun
is              AUX            auxiliary
also            ADV            adverb
the             DET            determiner
CEO             NOUN           noun
of              ADP            adposition
Tesla           PROPN          proper noun
.               PUNCT          punctuation

                SPACE          space
The             DET            determiner
company         NOUN           noun
headquarters    NOUN           noun
is              AUX            auxiliary
located         VERB           verb
in              ADP            adposition
Texas           PROPN          proper noun
.               PUNCT          punctuation

                SPACE          space
On              ADP            adposition
July            PROPN          proper noun
15              NUM            numeral
,               PUNCT          punctuation
2025            NUM            numeral
,               PUNCT          punctuation
Tesla           PROPN          proper noun
announced       VERB           verb
a               DET            determiner
new             ADJ            adjective
AI              PROPN          proper noun
model           NOUN           noun
.               PUNCT          punctuation

                SPACE          space
```

Step 5: Named Entity Recognition (NER)

NER identifies real-world entities such as:

- Person
- Organization
- Date
- Time
- Money
- Location
- Country

```
print(f"{'Entity':20} {'Label':15} {'Meaning'}")

print("-"*70)

for ent in doc.ents:
    print(f"{ent.text:20} {ent.label_:15} {spacy.explain(ent.label_)}")
```

```
Entity               Label          Meaning
----------------------------------------------------------------------
Elon Musk            PERSON         People, including fictional
2002                 DATE           Absolute or relative dates or periods
Tesla                ORG            Companies, agencies, institutions, etc.
Texas                GPE            Countries, cities, states
July 15, 2025        DATE           Absolute or relative dates or periods
Tesla                ORG            Companies, agencies, institutions, etc.
AI                   GPE            Countries, cities, states
```

Step 6: Entity Validation

Entity validation checks whether an entity has been correctly identified along with its label.

```
for ent in doc.ents:
    print(f"{ent.text} --> {ent.label_} ({spacy.explain(ent.label_)})")
```

```
Elon Musk --> PERSON (People, including fictional)
2002 --> DATE (Absolute or relative dates or periods)
Tesla --> ORG (Companies, agencies, institutions, etc.)
Texas --> GPE (Countries, cities, states)
```

```
July 15, 2025 --> DATE (Absolute or relative dates or periods)
Tesla --> ORG (Companies, agencies, institutions, etc.)
AI --> GPE (Countries, cities, states)
```

Step 7: Dependency Parsing (Optional)

Dependency Parsing shows the grammatical relationship between words in a sentence.

```
print(f"{'Token':15} {'Dependency':15} {'Head'}")

print("-"*45)

for token in doc:
    print(f"{token.text:15} {token.dep_:15} {token.head.text}")
```

```
Token           Dependency      Head
---------------------------------------------

                dep             Elon
Elon             compound        Musk
Musk             nsubj           founded
founded          ROOT            founded
SpaceX           dobj            founded
in               prep            founded
2002             pobj            in
.                punct           founded

                dep             .
He               nsubj           is
is               ROOT            is
also             advmod          is
the              det             CEO
CEO              attr            is
of               prep            CEO
Tesla            pobj            of
.                punct           is

                dep             .
The              det             headquarters
company          compound        headquarters
headquarters     nsubjpass       located
is               auxpass         located
located          ROOT            located
in               prep            located
Texas            pobj            in
.                punct           located

                dep             .
On               prep            announced
July             pobj            On
15               nummod          July
,                punct           July
2025             nummod          July
,                punct           announced
Tesla            nsubj           announced
announced        ROOT            announced
a                det             model
new              amod            model
AI               compound        model
model            dobj            announced
.                punct           announced

                dep             .
```

Step 8: Visualizing Named Entities

spaCy provides a built-in visualization tool called displacy.

```
from spacy import displacy
displacy.render(doc, style="ent", jupyter=True)
```

Elon Musk **PERSON** founded SpaceX in 2002 **DATE** .

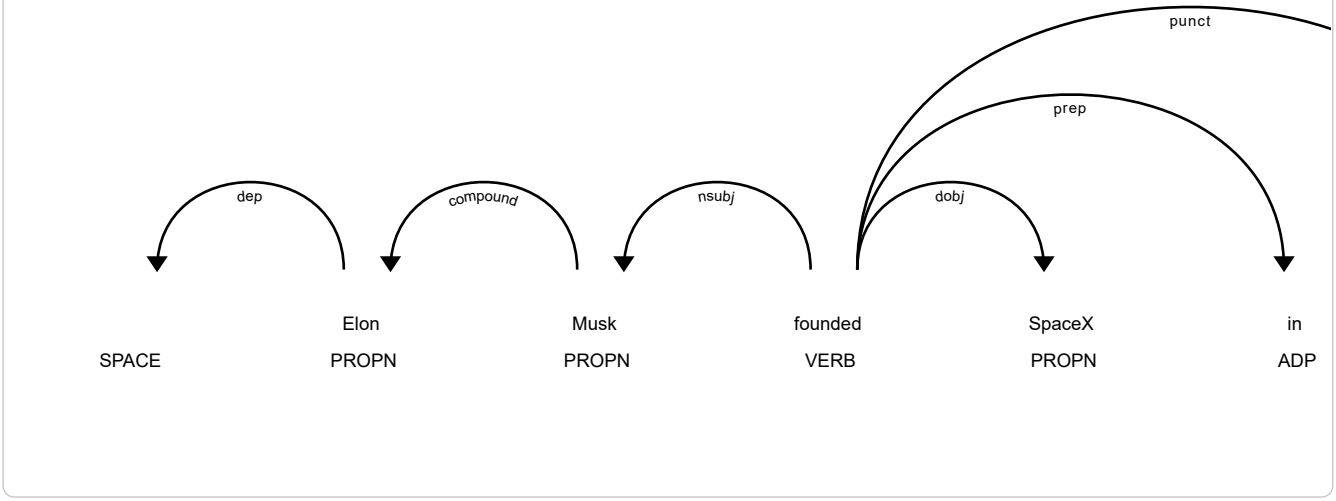
He is also the CEO of Tesla **ORG** .

The company headquarters is located in Texas **GPE** .

On July 15, 2025 **DATE** , Tesla **ORG** announced a new AI **GPE** model.

Step 9: Visualizing Dependency Tree

```
displacy.render(doc, style="dep", jupyter=True)
```



Conclusion

In this practical, we successfully implemented an introductory NLP pipeline using spaCy.

The pipeline included:

- Text Processing
- Sentence Segmentation
- Tokenization
- POS Tagging
- Named Entity Recognition
- Entity Validation
- Dependency Parsing

This practical demonstrates the basic workflow of Natural Language Processing and serves as the foundation for advanced NLP applications such as sentiment analysis, text summarization, and conversational AI.